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ORIGINAL RESEARCH

Predicting Climate Change Impacts on Temperature and Rainfall in Bangladesh Using Ensemble Learning: A Statistical Yearbook Analysis

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Abstract

Climate change in Bangladesh is causing unstable temperatures and rainfall, increasing the risk of floods, droughts, and agricultural disruption. Previous studies missed integrating temperature and rainfall data with feature selection and Explainable AI (XAI). This paper introduces a novel approach that integrates machine learning (ML) and deep learning (DL) methods with feature selection and XAI to improve climate impact predictions across the region. Using a comprehensive dataset from the Bangladesh Statistical Yearbook (1981–2023), our feature analysis highlights that key months such as April, February, and June strongly influence temperature and rainfall predictions, while location plays a greater role in shaping rainfall patterns than temperature. Our proposed ensemble model outperformed traditional and previous models across three climate datasets, achieving R^2 scores of 1.00, 0.79, and 0.99 for maximum temperature, minimum temperature, and rainfall, respectively. It also recorded low MAE values of 0.08, 0.39, and 3.46, and RMSE values of 0.10, 0.50, and 4.20 correspondingly. XAI techniques like LIME improved prediction interpretability by revealing key factors influencing the model's decisions. Critical influences included lower March and July temperatures for maximum temperature predictions, while higher January and February temperatures positively affected the model. For rainfall, significant contributions from July and September rainfall were observed, while lower rainfall in April and December negatively impacted predictions. These insights support better agriculture and water management by identifying key climate drivers. Future work can enhance the model by adding more meteorological variables and considering large-scale climate patterns for deeper regional insights.

Keywords Climate change · Explainable AI · Deep learning · Temperature prediction · Rainfall prediction

Introduction

Climate change represents one of the most pressing global challenges of the twentieth century, with widespread consequences for weather patterns, ecosystems, and human livelihoods. The Intergovernmental Panel on Climate Change (IPCC) has warned that rising global temperatures and altered precipitation patterns pose serious threats to biodiversity, food security, and agricultural productivity, particularly in regions like South Asia, which are highly vulnerable due to their geographic and socio-economic conditions [49]. Bangladesh, given its low-lying geography, high population density, and reliance on agriculture, is especially susceptible to climate variability. Increasing temperatures, erratic rainfall, and the growing frequency of extreme weather events such as heatwaves, floods, droughts, and tropical cyclones place immense pressure on the country's agricultural sector,

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food security, and public health systems. Understanding these impacts is essential for formulating effective adaptation strategies to safeguard the livelihoods of millions of Bangladeshis [24].

Bangladesh's agriculture is increasingly threatened by more frequent floods, droughts, heatwaves, and storms driven by climate change. Variability in temperature and rainfall has led to significant fluctuations in crop yields, exacerbating food insecurity in the region [20]. It is predicted that global warming of 1.5°C compared to 2°C could reduce mean sea level rise by 0.1 meters by 2100, with sea levels continuing to rise well beyond 2100 depending on future emission pathways [37]. Furthermore, studies show that Bangladesh is among the countries most at risk from these climate-induced challenges [26]. The country's agricultural yields may decrease by as much as 30%, significantly higher than the global average, if climate conditions continue to deteriorate [39]. Bangladesh's growing fisheries sector is also at risk, as climate change makes it harder for farmers to understand and adapt to shifting weather patterns [8].

Urbanization in Bangladesh makes climate impacts worse by changing natural landscapes, causing temperature differences between cities and villages, and making extreme weather more harmful [22]. These challenges, combined with climate change, affect agriculture and food production [45]. Accurate predictions of temperature and rainfall are essential to help policymakers plan and reduce risks to food security, health, and the economy [16, 17, 46].

However, despite the critical need for accurate forecasting, there remain notable gaps in climate prediction models in Bangladesh–particularly in the integration of both temperature and rainfall data using advanced analytical techniques. Several previous studies have relied on traditional statistical models or basic machine learning (ML) techniques without optimizing for feature co-linearity or incorporating deep learning (DL) architectures for improved prediction accuracy [9, 34]. This study addresses these gaps by developing an ensemble model that integrates both ML and DL methods to improve the precision of temperature and rainfall forecasts. Using data from the Bangladesh Statistical Yearbook and employing Explainable Artificial Intelligence (XAI) techniques, this study aims to provide insights into the key factors driving climate predictions. By enhancing model transparency, this approach builds trust among stakeholders, enabling them to make informed decisions about future climate adaptation strategies.

The key contributions of this study are as follows:

- To develop an ensemble model that combines ML and DL for precise temperature and rainfall predictions in Bangladesh using extensive historical data.
- The Principal Component Analysis and Information Gain (PCA-IG) based feature selection method effectively identified the most influential features, enabling deeper analysis and revealing which months had the greatest impact on the prediction outcomes.
- To utilize XAI methods like LIME for analyzing climatic factors' importance, improving model clarity and aiding stakeholders in interpretation.
- Developed an ensemble modeling framework that integrates decades-long temperature and rainfall data, offering improved regional climate predictions for Bangladesh compared to existing approaches.

This article is organized as follows: "Related Works" reviews relevant literature; "Methodology" outlines the proposed methodology, encompassing feature analysis, model training, explainable AI techniques, evaluation metrics, and the experimental setup; "Experimental Results and Analysis" presents the results with a thorough analysis; "Discussion" discusses the implications of the findings; and "Conclusion and Future Work" concludes with future directions for enhancing climate factors in Bangladesh.

Related Works

Bangladesh's climate is changing dramatically, with increased frequency of extreme weather events, changed precipitation patterns, and rising temperatures. Between 1901 and 2020, Bangladesh saw a noticeable warming trend, with average temperatures rising by about 2°C [29]. The region's livelihoods in general, as well as agriculture and health, are seriously threatened by these trends. There are now more health concerns associated with heat exposure due to the yearly average wet bulb globe temperature (WBGT) rising by 0.08–0.5 °C every decade [30]. Extreme weather events, such heat waves and droughts, are occurring more frequently, which has serious effects on agriculture and food security [25]. Despite these concerning tendencies, some research indicates that Bangladesh may be able to lessen the negative consequences of climate change through the use of adaptation tactics and climate-resilient agricultural techniques [28, 36].

Bangladesh's agriculture sector and overall climate resilience are increasingly threatened by climate-induced temperature fluctuations. From 1949 to 2013, the country's average annual temperature rose by approximately 0.13° C per decade, with seasonal disparities [5]. Extreme temperature events, such as heat waves and cold waves, have become more frequent, especially in areas like Jashore [47]. Climate projections estimate that summer temperatures may increase by 3.24°C to 5.77°C by the end of the century, severely affecting crop yields and food security [10]. Spatial analysis reveals that the most pronounced warming trends are occurring in the southern and southeastern regions.

The detection and forecast of rainfall is becoming a significant issue in many areas as a result of global warming. The data represents district-specific rainfall measurements (in mm) across India for the years 1951 to 2015. The study conducted by Arnav Garg and Kanchipuram [21] examines three ML algorithms: Support Vector Machine (SVM), Support Vector Regression (SVR), and K-Nearest Neighbor (KNN), applied to patterns of rainfall within a year. Among these algorithms, the SVM algorithm was found to perform the best. However, the research did not specify which environmental features significantly impact rainfall intensity. This paper identifies the environmental features that have both positive and negative impacts on rainfall and uses these features to predict the daily rainfall amount.

By applying ML and DL techniques, rainfall prediction has made tremendous progress towards more accurate forecasts [18, 33]. Logistic regression, ARIMA, SVM, and Artificial Neural Networks (ANN) are among the frequently utilized methods [11]. These models forecast rainfall patterns by analyzing historical data. Although the accuracy of rainfall predictions can be improved with the use of ML and DL approaches, generalization of the models remains difficult, and constant data updates are required to adjust to shifting weather patterns. To capture complex interactions between several meteorological indicators and improve the accuracy of rainfall forecasts, neural networks are utilized [13].

Duan, H. et al. [19] utilized KNN, Random Forest, and SVM algorithms for climate prediction, achieving accuracies of 82.4%, 85.6%, and 81.4%, respectively. Although Random Forest demonstrated the highest predictive accuracy, the substantial variation in F1-scores highlighted challenges related to class balance and overall model reliability. Liyew et al. [35] compared Multiple Linear Regression, Random Forest, and XGBoost for predicting daily rainfall over a 20-year period. XGBoost emerged as the best-performing model, although the study did not incorporate sensor data and relied on a limited feature set. This restricts the model's scalability across diverse geographies.

Forecasting temperatures is crucial for both agricultural planning and the stability of infrastructure [50]. Milad, Abdalrhman, et al. [38] examines how air temperature at various depths and times predicts asphalt pavement temperature and assesses performance. The data for this study was collected over a period from March 2012 to February 2013. Temperature measurements were taken at various depths, specifically at 0, 2, 5.5, and 7 cm. The research paper compares the effectiveness of four DL prediction methods: Bidirectional Long Short-Term Memory (Bi-LSTM), Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU). Among these, Bi-LSTM outperformed the others, achieving an R^2 value of 0.9555. This study highlights the potential of DL in temperature prediction. It is noted that previous models often produced inaccurate data for temperature variations.

Similarly, Gong et al. [23] utilized the ERA5 dataset alongside deep learning techniques such as CNN, ConvLSTM, and SAVP to estimate 2-meter diurnal temperatures. Their findings indicated that SAVP performed best for short-term forecasts of up to 12 h. However, CNNs on their own proved inadequate for extended predictions due to their inability to capture temporal dependencies, resulting in accumulating errors within a few hours. While these studies represent meaningful progress, their limitations in adaptability, regional applicability, and feature clarity underscore the need for more versatile and robust models. Azeem et al. [9] integrating multiple ML models, this method reached an accuracy of 86.5%, demonstrating the effectiveness of ensemble techniques in enhancing predictive performance.

Recent research has increasingly focused on incorporating multiple climate variables for improved forecasting. For instance, Random Forest models have demonstrated rainfall prediction accuracies of up to 83.2% by utilizing diverse meteorological parameters [4]. Similarly, Kothai et al. [32] found that integrating seasonal factors such as humidity and wind speed significantly enhances prediction accuracy. Studies reveal certain seasonal trends in temperature and precipitation, with the highest levels of precipitation usually falling between March and May. Although ML greatly improves temperature and rainfall forecasts. By applying Synthetic Minority Oversampling Technique (SMOTE), the distribution becomes more balanced, improving the accuracy of analysis and predictions [34].

XAI methods have been introduced to mitigate the black-box nature of ML models[14]. LSTM-based approaches have demonstrated rainfall prediction accuracies of up to 92.2% [31], yet tools like Layer-wise Relevance Propagation (LRP) and Integrated Gradients are crucial for result interpretation. Patel et al. [43] highlighted that attention mechanisms can pinpoint the most influential features, supporting resource management strategies. However, ensuring the robustness and applicability of these techniques across diverse climate conditions remains a challenge, particularly in areas with sparse meteorological data. The XAI process is structured around the feature importance criterion, ensuring a systematic selection of explainability techniques and a thorough evaluation of explanation consistency. This approach helps refine model transparency by identifying the most relevant features and validating the reliability of the interpretations provided [27]. However, XAI methods like

Layer-wise Relevance Propagation often fail in heterogeneous climates due to data drift [12].

Although numerous studies have leveraged ML and DL techniques for rainfall and temperature prediction. Most existing climate-forecasting studies use limited features, rely on opaque "black-box" models, and stick to single algorithms–making them less accurate and hard to generalize across Bangladesh's varied zones. Furthermore, explainability is frequently overlooked, making it difficult to interpret predictions and apply them in real-world decision-making. To address these gaps, we propose a new ensemble framework that uses diverse meteorological and environmental inputs and integrates XAI to improve accuracy, transferability, and transparency. Table 1 presents a summarized overview of existing climate-related studies reviewed in the literature.

Methodology

In this section, we describe the approach used in our study to predict the standard average maximum and minimum temperatures, as well as average rainfall. Our method involves several key steps: gathering and preparing the data, feature engineering, selecting the right model, and incorporating explainable AI. Figure 1 offers a visual overview of our proposed approach.

Data Collection and Pre-processing

The dataset used in this study was collected from a government website and derived from the Statistical Yearbook of Bangladesh for the years 2011–2023 [40]. This yearbook provides important data on various aspects of Bangladesh, such as land, environment, population, agriculture, and fisheries. Our focus, however, was on monthly records of the standard maximum and minimum average temperatures, as well as average rainfall, from 1981 to 2023. The data for 1981–2010 was provided as a single set, while the data from 2011 to 2023 was available separately for each year. The dataset covers 42–48 weather stations across Bangladesh, including major locations like Dhaka, Chittagong, Cox's Bazar, Rajshahi, Dinajpur, Khulna, Teknaf, and other areas. We gathered all this information to create a large, comprehensive dataset for analyzing our study's outcomes. Additionally, for testing our model, we used three independent datasets for our three target outcomes: one for standard average maximum temperature, one for standard average minimum temperature, and one for average rainfall. These

Table 1 Summary of related works

References	Target	Method	Best performance	Limitation/critique
[4]	Temp. and rainfall prediction	NB, ANN, LogitBoost, RIPPER, DS, AdaBoost, RF	Accuracy of 83.2%	Limited discussion of how weather features affect prediction reliability
[35]	Rainfall Prediction	MLR, RF, XGBoost	MAE = 3.58, RMSE = 7.85	Excludes sensor data and uses limited environmental features
[38]	Temperature Prediction	Bi-LSTM, CNN, LSTM, GRU	$R^2 = 0.9555$	Previous models produced unreliable temperature variation outputs
[31]	Rainfall Prediction	XGBoost, RF, SVM, MLR, LSTM	Accuracy = 92.2%, MAE = 11.7%	Data quality issues may limit robustness despite preprocessing
[19]	Rainfall Prediction	KNN, SVM-Random forest	Accuracies respectively: 82.4464%, 85.6416%, and 81.3915%	The paper examines the shortcomings of traditional empirical and physical statistical approaches in rainfall prediction
[34]	Rainfall Prediction	RF, KNN, DT	F1-score =0.91 Precision =0.89Recall=0.94Accuracy =0.91	The paper lacks information about the criteria or methods used to choose features for model training
[44]	Rainfall prediction	RF, SVM, RNN, LSTM	Accuracy = 94.68%	The paper does not specify the types of machine learning models used or their limitations
[9]	Rainfall prediction	SVM, Neural Networks, ANN and Voting Ensemble	Accuracy = 86.5%	Emphasis on short-term forecasting may limit its usefulness for agricultural planning and disaster preparedness

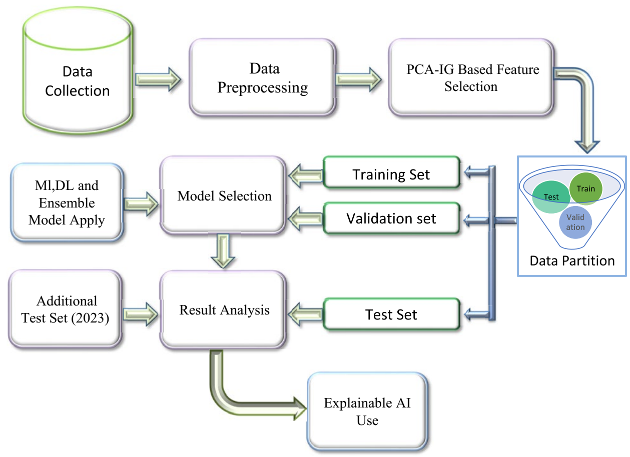


Fig. 1 Methodological overview for predicting average maximum and minimum temperatures and rainfall in Bangladesh

Table 2 Train, test, and validation split of the employed datasets

Dataset name	Target column	Number of samples	Train	Test	Valid
Average maximum temperature	Year	431	276	86	69
Average minimum temperature	Year	426	273	85	68
Average rainfall	Year	433	277	87	69

datasets were also taken from the Statistical Yearbook of Bangladesh (43rd edition) [41].

Once the data collection was complete, we moved on to the data preprocessing step. Data preprocessing in ML involves cleaning, transforming, and organizing raw data to improve model accuracy and performance. In our dataset, there were some missing values, which we handled using the median imputation method. The dataset consists of 14 columns, one of which is categorical: the ‘Station’ column. We applied label encoding to convert this categorical data into numerical values. The dataset includes 431 data points for the standard average maximum temperature, 426 for the standard average minimum temperature, and 433 for average rainfall. After resolving all data-related issues, we experimented with different data splitting strategies, including train-test and train-test-validation setups. We found that using 80% of the data for training, 20% for testing, and reserving 20% of the training data for validation gave the best results. This partitioning helped ensure the data was well-prepared for accurate analysis and predictive modeling. Table 2 presents the specific breakdown of the data for model training. As shown in Table 2, the target column for all datasets is ‘Year’, which represents the annual average of monthly values for standard maximum temperature, standard minimum temperature, and total rainfall.

Feature Engineering

We proceeded to feature engineering after finishing data preprocessing. Feature engineering is the process of identifying and refining the most valuable data points to enhance a model’s accuracy and readability [6]. Feature engineering involves refining raw data to enhance the performance and accuracy of ML models. This study [42] explored various feature selection techniques and found that the Principal Component Analysis – Information Gain (PCA-IG) method most effectively identified the key features. It also enhanced model performance, making it the chosen approach for feature selection in our analysis. This is a hybrid technique that combines dimensionality reduction with an entropy-based measure to identify the most relevant features for the ML task. PCA helps eliminate redundant or highly correlated features by concentrating on the most significant components. It first transforms the original feature space into principal components using the PCA equation.

$$Z_{reduced} = ZW \tag{1}$$

where, the standardized data matrix Z (zero mean, unit variance) is transformed using the eigenvector matrix W to produce Z_{reduced}, which represents the dataset in a lower-dimensional space.

After obtaining the principal components, calculate the IG for each component regarding the target variable Y. The IG quantifies the reduction in uncertainty about Y when a specific principal component X_{PC} is known, as illustrated in Eq. 2.

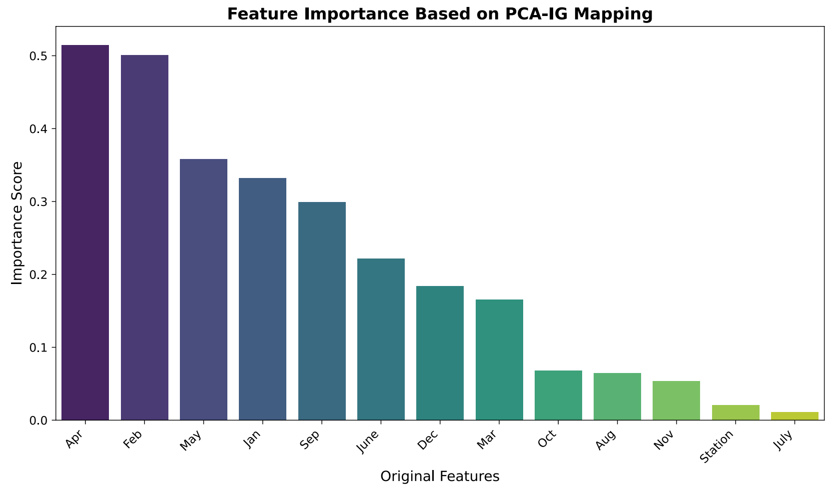
$$IG(Y, X_{\text{PC}}) = H(Y) - H(Y \mid X_{\text{PC}}) \tag{2}$$

In this context, $H(Y)$ represents the entropy of the target variable Y, while $H(Y|X_{\text{PC}})$ denotes the conditional entropy of Y given the principal component X_{PC}.

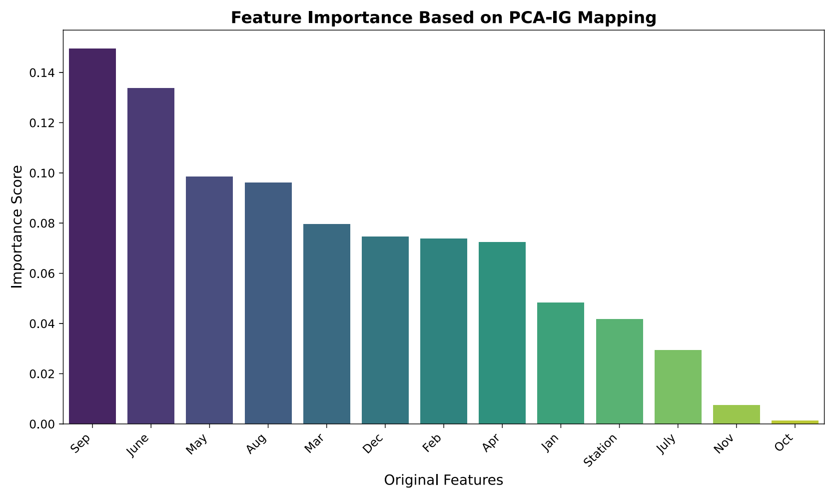
So, we conducted an analysis of three datasets utilizing the PCA-IG based feature selection method, as depicted in Fig. 2.

Figure 2 shows the feature importance scores derived from the PCA-IG technique, where principal components are mapped back to the original features. In Fig. 2a, from the maximum temperature dataset analysis, the most significant feature is ‘April’, with an importance score of 0.514, followed by ‘February’, which has a score of 0.501. Other key features include ‘May’ (0.359), ‘January’ (0.333), and ‘September’ (0.300). Lower-ranking features like ‘July’ (0.011) and ’Station’ (0.020) show minimal importance in predicting the target variable. These scores highlight the key months for the model.

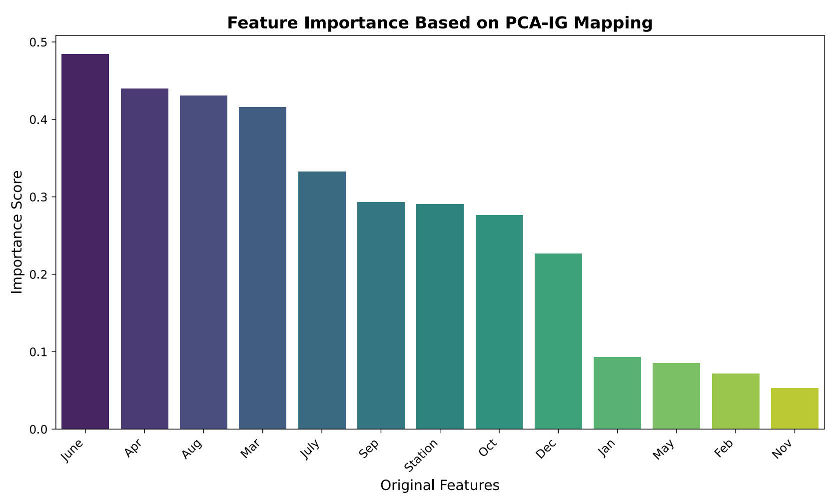
Fig. 2 Bar plot of PCA-IG based feature importance for maximum temperature, minimum temperature, and rainfall



(a) Maximum Temperature Feature Importance.



(b) Minimum Temperature Feature Importance.



(c) Rainfall Feature Importance.

In Fig. 2b, from the minimum temperature dataset analysis, the bar chart shows ‘September’ as the most significant feature with a score of 0.149, followed by ‘June’ (0.134). ‘May’ (0.099) and ‘August’ (0.096) also have notable importance, while ‘November’ (0.010) and ’October’ (0.003) contribute minimally. These scores show the varying importance of different months in predicting the target variable. Higher importance scores for months like September and June suggest they have a stronger impact on model performance.

In Fig. 2c, from the average rainfall dataset analysis, ‘June’ is the most influential feature with a score of 0.485, followed by ‘April’ (0.440), ‘August’ (0.430), and ‘March’ (0.416), highlighting the importance of summer and spring months. In contrast, ‘November’ (0.053) and ‘February’ (0.071) have minimal influence, while the ‘Station’ feature, representing location, holds moderate significance at 0.291. Findings suggest that months like June and April significantly contribute to the model, while features like November have less impact. This analysis can help refine the model by focusing on high-impact features to enhance performance.

A cross-validation technique was employed to derive the feature importance scores. In these feature importance figures, each feature represents a month, corresponding to the average standardized maximum temperature, average standardized minimum temperature, and average monthly rainfall for that month. The feature importance results varied across datasets, indicating that different months or seasons have varying impacts under different conditions. This suggests that climatic factors influence each dataset uniquely depending on the time of year.

Machine Learning and Deep Learning Model Selection

After completing the data cleaning, splitting, and feature selection processes, we proceeded to evaluate and select the best ML and DL models for our three datasets. We employed several models in our analysis and present in this paper only the top five performing ML models, one DL model, and our proposed ensemble model for prediction. All ML models were built using the scikit-learn library in Python, while the ANN model was developed and trained using TensorFlow/Keras. This approach ensured a consistent setup for evaluating and comparing the performance of all models across different techniques. A detailed description of all the models used in this study is provided below.

Random Forest Regression (RFR)

RFR is a powerful and widely used ML technique that works by building many decision trees and averaging their results to make predictions. It reduces overfitting by combining the results of diverse trees, each trained on a random subset of the data and features, thereby improving generalization and robustness. RFR is particularly effective in handling nonlinear relationships and high-dimensional datasets.

Linear Regression (LR)

LR is a simple and widely used method that finds the straight-line relationship between input variables and an outcome. It works by estimating the best-fitting line that predicts the outcome based on the inputs, making it easy to understand and interpret how different factors affect the result.

Lasso with Cross-Validation Regression (LassoCV)

LassoCV is a version of Lasso Regression that automatically finds the best level of regularization using cross-validation. It helps simplify the model by reducing less important feature weights to zero, making the results more accurate and easier to interpret.

K-Nearest Neighbors (KNN) Regression

KNN Regression makes predictions by looking at the closest data points around a given sample and averaging their values. It’s a straightforward and intuitive approach that works well when similar inputs tend to have similar outcomes.

Huber Regression

Huber Regression is a flexible approach that handles outliers better than traditional methods. It treats small errors normally but reduces the impact of large errors, helping the model stay accurate even when the data has unusual or extreme values.

Artificial Neural Network (ANN)

ANN is a DL model inspired by the human brain that learns complex patterns through interconnected layers of neurons. To boost performance, we first scaled the data. The network included an input layer, two hidden layers with dropout to help prevent overfitting, and an output layer. This setup allowed the model to learn complex patterns in the data while still making reliable and generalizable predictions.

We proposed an ensemble model that demonstrated superior performance compared to the other models, as detailed in the following section.

Proposed Ensemble Learning

Our proposed ensemble combines Huber Regressor and ANN using weighted averaging to leverage their complementary strengths–Huber Regressor's robustness to outliers and ANN's ability to capture complex nonlinear patterns–enhancing overall accuracy and generalizability. This method is commonly referred to as model stacking or weighted averaging. The Huber Regressor is a robust regression model that minimizes the Huber loss function, which is less sensitive to outliers than the standard least squares loss. The Huber loss is defined as:

$$L_\delta(y, f(x)) = \begin{cases} \frac{1}{2}(y - f(x))^2 & \text{if } |y - f(x)| \leq \delta \\ \delta \cdot (|y - f(x)| - \frac{1}{2}\delta) & \text{otherwise} \end{cases} \tag{3}$$

where y is the true value and $f(x)$ is the predicted value. The parameter δ separates small errors from large ones.

ANN comprises an input layer that accepts scaled features, two hidden layers with 128 and 64 neurons utilizing the ReLU activation function, and dropout layers to mitigate overfitting by randomly omitting neurons during training. The output layer consists of a single neuron with a linear activation function, generating the final prediction. The overall operation of the ANN can be represented by the equation for the final output:

$$\hat{y} = W_{\text{out}} \cdot H + b_{\text{out}} \tag{4}$$

where H is the output from the last hidden layer, W_{out} is the weight matrix for the output layer, and b_{out} is the bias for the output layer.

The ensemble model utilizes a weighted average of the predictions from both models, allowing it to harness the strengths of each approach. This combination is particularly useful for improving robustness against outliers. The ensemble prediction is expressed mathematically as follows:

$$\hat{y}_{\text{ensemble}} = w \cdot \hat{y}_{\text{Huber}} + (1 - w) \cdot \hat{y}_{\text{ANN}} \tag{5}$$

where $\hat{y}_{\text{ensemble}}$ represents the final ensemble prediction, $\hat{y}_{\text{Huber}}$ is the prediction from the Huber Regressor, and $\hat{y}_{\text{ANN}}$ is the prediction from the ANN. The weight parameter w (with $0 < w < 1$) determines the contribution of the Huber Regressor to the final prediction.

Algorithm 1 outlines our proposed ensemble model, which combines a Huber Regressor and an ANN for more accurate predictions. It starts by standardizing the data, then trains both models using the Huber loss to handle outliers effectively. The ANN uses three hidden layers and the Adam optimizer, while the Huber Regressor is trained in parallel. After both models make predictions, their outputs are combined using weighted averaging.

Input: Training data $\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}}$, Test data $\mathbf{X}_{\text{test}}, \mathbf{y}_{\text{test}}$
Initialize: Huber parameter $\delta = 1.0$, learning rate $\alpha = 0.0005$, ensemble weight $w = 0.5$
procedure FEATURE SCALING
  Standardize $\mathbf{X}_{\text{train}}, \mathbf{X}_{\text{test}}, \mathbf{y}_{\text{train}}, \mathbf{y}_{\text{test}}$
end procedure
procedure HUBER LOSS FUNCTION
  $L_\delta(y_{\text{true}}, y_{\text{pred}}) = \begin{cases} \frac{1}{2}(y_{\text{true}} - y_{\text{pred}})^2, & \text{if } |y_{\text{true}} - y_{\text{pred}}| \leq \delta \\ \delta \cdot (|y_{\text{true}} - y_{\text{pred}}| - \frac{\delta}{2}), & \text{otherwise} \end{cases}$
end procedure
procedure TRAIN ANN
  Build an ANN with the following layers: $128 \to 64 \to 32 \to 1$
  Compile the ANN with Adam optimizer, learning rate α, and Huber loss
  Train the model on scaled $\mathbf{X}_{\text{train}}$ and $\mathbf{y}_{\text{train}}$
end procedure
procedure PREDICTIONS (ANN)
  Predict $\hat{y}_{\text{ann}}$ using the trained ANN on $\mathbf{X}_{\text{test}}$
  Inverse scale $\hat{y}_{\text{ann}}$ to return it to the original scale
end procedure
procedure HUBER REGRESSOR
  Fit the Huber Regressor to the scaled training data $\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}}$
  Predict $\hat{y}_{\text{huber}}$ on $\mathbf{X}_{\text{test}}$
end procedure
procedure ENSEMBLE MODEL
  Compute the ensemble prediction: $\hat{y}_{\text{ensemble}} = w \cdot \hat{y}_{\text{huber}} + (1 - w) \cdot \hat{y}_{\text{ann}}$
end procedure

Algorithm 1 Hybrid Huber Regressor and ANN Model

Table 3 Hyper-parameter settings for various models utilized in this study

Model name	Hyper-parameter setting
RFR	`n_estimators` = 100, `random_state` = 42, `max_depth` = None
LR	`fit_intercept` = True, `normalize` = False, `copy_X` = True
LassoCV	`cv` = 5, `random_state` = None, `max_iter` = 1000
KNN	`n_neighbors` = 5, `weights` = 'uniform', `algorithm` = 'auto'
Huber	`epsilon` = 1.35, `max_iter` = 1000, `alpha` = 0.0
ANN	`hidden_layers` = [128, 64, 32], `dropout_rate` = 0.2, `learning_rate` = 0.0005, `batch_size` = 16, `epochs` = 300–500, `validation_split` = 0.2
Proposed Model (Ensemble)	`weight` = 0.5, `model_1` (Huber Regressor) hyperparameters = [`epsilon` = 1.35], `model_2` (Improved ANN) hyperparameters = [`hidden_layers` = [31], `dropout_rate` = 0.2, `learning_rate` = 0.0005, `batch_size` = 16, `epochs` = 300–500]

Hyper-Parameter Tuning

To select the best model for predicting temperature and rainfall, we employed various ML, DL, and ensemble models for analysis. Each model's hyperparameters were fine-tuned and tested with selected features to identify the optimal configuration. In our study, we used manual hyperparameter tuning based on domain knowledge and iterative trials. Due to computational limits and dataset specifics, we did not apply automated methods like grid or random search, as shown in Table 3.

Table 3 summarizes key hyperparameters: for RFR, $n_{\text{estimators}}$ set to 100 improves prediction, random_state ensures reproducibility, and max_depth set to None allows full tree growth. For LR, fit_intercept set to True adds an intercept, while copy_X set to True preserves input data. For LassoCV, cv set to 5 controls cross-validation folds, and max_iter of 1000 ensures convergence. In KNN, $n_{\text{neighbors}}$ is 5, weights is uniform, and algorithm is auto. For Huber Regressor, ϵ is 1.35 for outlier handling, max_iter is 1000, and α is 0.0 indicating no regularization. The ANN model has three hidden layers (128, 64, 32 neurons) with ReLU activation and dropout of 0.2 after the first two layers. It uses Adam optimizer (learning rate 0.0005), trains for 300–500 epochs, batch size 16, with a validation split of 0.2. The proposed model combines a Huber Regressor (epsilon 1.35, 1000 max iterations) and an improved ANN with three hidden layers (128, 64, 32 neurons), ReLU activation, and 0.2 dropout. The ANN uses Adam optimizer (learning rate 0.0005), trained for 300 epochs on minimum temperature data and 500 epochs on maximum temperature and rainfall data, with batch size 16 and 0.2 validation split. The ensemble balances both models equally with a weight of 0.5 to enhance performance.

Evaluation Matrices

Following the selection of models and the tuning of hyperparameters, we assessed their performance through six essential regression metrics. These metrics include Mean Absolute Error (MAE), which measures the average absolute difference between predicted and actual values; Mean Squared Error (MSE), which calculates the average squared difference; and Root Mean Squared Error (RMSE), offering an interpretable magnitude of error. We also assessed Coefficient of Determination (R^2), which indicates the proportion of variance explained by the model; Root Mean Squared Logarithmic Error (RMSLE), which evaluates the ratio of predicted to actual values; and Mean Absolute Percentage Error (MAPE), representing the average percentage error between predictions and actual outcomes. Optimal performance is indicated by lower values for MAE, MSE, RMSE, and RMSLE, while higher (R^2) values and lower MAPE percentages reflect better model performance. The equations for these evaluation metrics are detailed below.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{6}$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{7}$$

$$RMSE = \sqrt{MSE} \tag{8}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y}_i)^2} \tag{9}$$

$$RMSLE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\log(1+y_i) - \log(1+\hat{y}_i))^2} \tag{10}$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100 \tag{11}$$

In this context, N refers to the total number of samples, y_i represents the actual values, $\hat{y}_i$ denotes the predicted values, and $\bar{y}_i$ indicates the mean of the actual values.

Table 4 Prediction results of standard average maximum temperature from all models employed

Model	MAE	MSE	RMSE	R^2	RMSLE	MAPE (%)
RFR	0.44	6.66	2.58	0.14	0.01	1.04
LR	0.13	0.16	0.39	0.98	0.01	0.36
LassoCV	0.12	0.32	0.57	0.96	0.00	0.31
KNN regressor	0.61	6.88	2.62	0.11	0.06	1.56
Huber regressor	0.09	0.06	0.24	0.99	0.01	0.26
ANN	0.16	0.06	0.24	0.99	0.01	0.52
Proposed ensemble model	**0.08**	**0.02**	**0.15**	**1.00**	**0.00**	**0.25**

Table 5 Prediction results of standard average minimum temperature from all models employed

Model	MAE	MSE	RMSE	R^2	RMSLE	MAPE (%)
RFR	0.64	1.76	1.33	−0.30	0.00	2.87
LR	0.53	0.77	0.88	0.43	0.00	2.41
LassoCV	0.74	1.41	1.19	−0.04	0.01	3.41
KNN Regressor	0.55	0.92	0.96	0.32	0.04	2.46
Huber Regressor	0.39	0.37	0.61	0.73	0.00	1.79
ANN Model	0.51	0.50	0.71	0.63	0.03	2.37
Proposed Ensemble Model	**0.39**	**0.29**	**0.53**	**0.79**	**0.02**	**1.80**

Explainable Artificial Intelligence (XAI)

The purpose of XAI is to help users understand the factors that influence results and the reasoning behind ML models' conclusions. There are several XAI techniques available, such as SHAP, Shapash, and LIME. We chose LIME because it provides clear, focused explanations for individual predictions, which aligns with our goal of understanding results for specific stations or years [48]. Its straightforward, easy-to-visualize outputs make it especially helpful for climate policymakers and non-technical users [51, 52]. The details of the LIME technique are outlined below.

Local Interpretable Model-Agnostic Explanations (LIME)

LIME simplifies the interpretation of complex models by utilizing more straightforward models to clarify their behavior, thereby providing insights into the reasoning behind specific predictions[7]. The explanation model for LIME is as follows:

$$\hat{g}\left(x\right) = argmin_{g\epsilon G} L\left(f, g, \pi_{x'}\right) + \Omega\left(g\right) \quad (12)$$

In Eq. 12, the explanation model $\hat{g}(x)$ is designed to locally approximate the behavior of intricate models by leveraging a collection G of simpler models, typically linear in nature. A loss function $L(f, g, \pi_{x'})$ is employed to strike a balance between the model's fidelity and its simplicity, ensuring interpretability.

Experimental Setup

In this study, Google Colab served as our primary platform for designing, implementing, and training ML models [15]. The free edition of Google Colab provided a powerful and accessible computing environment with an NVIDIA Tesla T4 GPU, 12 GB of RAM, and a virtual CPU, allowing us to carry out extended experiments without requiring significant local computing resources. Its collaborative features enabled our research team to seamlessly share code and datasets, enhancing teamwork and overall productivity.

The ANN model employed was built using the TensorFlow library (version 2.x) [2], leveraging Keras as the high-level API for model construction and training. All code was developed in Python 3.8 [1]. Additionally, the Scikit-learn library (version 1.0.2) [3] was used for various ML tasks, including feature selection, regression, and evaluation.

All models were evaluated using 5-fold cross-validation to ensure reliable performance by reducing overfitting and maximizing the use of available data. Package versions and device specifications are documented to enhance reproducibility of the experiments.

Experimental Results and Analysis

This section presents the prediction results from various models, followed by a comparative analysis of the standard average maximum temperature, standard average minimum temperature, and average rainfall datasets. We also applied LIME to offer a deeper interpretation of the model predictions. Among the models tested, our proposed ensemble model demonstrated superior performance, outperforming

Table 6 Prediction results of average rainfall from all models employed

Model	MAE	MSE	RMSE	R^2	RMSLE	MAPE (%)
RFR	13.89	388.35	19.71	0.91	0.01	7.69
LR	3.65	38.20	6.18	0.99	0.03	2.02
LassoCV	4.30	35.18	5.93	0.99	0.03	2.32
KNN Regressor	12.73	372.66	19.30	0.92	0.08	6.41
Huber Regressor	5.29	54.31	7.37	0.99	0.04	2.86
ANN	6.66	75.58	8.69	0.98	0.05	3.81
Proposed Ensemble Model	**3.46**	**26.49**	**5.15**	**0.99**	**0.03**	**1.93**

all others. The following outlines the step-by-step approach used in this study.

Performance of Models

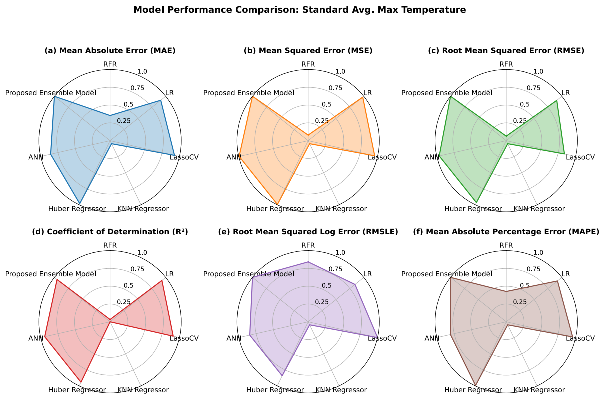
Now we compare the performance of different models: ML, DL, and our proposed ensemble model across each dataset using six evaluation metrics. Table 4 presents the results for the standard average maximum temperature dataset, Table 5 covers the standard average minimum temperature dataset, and Table 6 displays the outcomes for the average rainfall dataset, as shown below.

Table 4 presents clear differences in how well each model predicts the standard average maximum temperature, showcasing their unique strengths and weaknesses in performance. The Huber Regressor stands out with the lowest MAE of 0.09 and a high R-squared value of 0.99, indicating that it explains almost all the variance in the data. The Ensemble model, which combines the Huber Regressor and ANN, further enhances prediction performance with an even lower MAE of 0.08 and an R-squared of 1.00, demonstrating perfect predictive capability on the training set. The ANN model demonstrates exceptional performance, achieving a MAE of 0.16 and an R^2 of 0.99, indicating high model performance and minimal error in predictions. In contrast, the RFR, despite having a reasonable R-squared of 0.14, exhibits a significantly higher MAE of 0.44, indicating poorer performance in this context. LR and LassoCV also perform well, but their MAE values of 0.13 and 0.12, respectively, suggest that they are less effective than the Huber Regressor and the Ensemble model. Our proposed ensemble model outperforms all examined models in predicting the standard average maximum temperature, achieving the best performance metrics across the board.

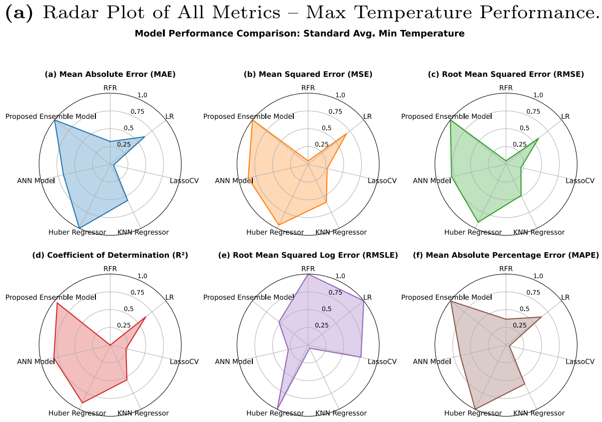
In Table 5, the results for standard average minimum temperature prediction reveal varying performances across the different models. The RFR, while powerful, exhibited a MAE of 0.64 and an R^2 value of −0.30, indicating a lack of predictive capability in this context. LR showed slightly better results with a MAE of 0.53 and an R^2 of 0.43, suggesting moderate performance. The LassoCV model performed less favorably with a MAE of 0.74 and a negative R^2, reflecting poor fit. KNN Regressor provided a more consistent prediction with a MAE of 0.55 and an R^2 of 0.32, although still below ideal performance. The Huber Regressor emerged as a strong contender, achieving a low MAE of 0.39 and a commendable R^2 of 0.73. The ANN model's MAE of 0.51 and R^2 of 0.63 indicated good performance but was surpassed by the Ensemble Model, which combined Huber and ANN methods, yielding the best results with a MAE of 0.39 and an impressive R^2 of 0.79. Therefore, the Ensemble Model is determined to be the best choice for predicting standard average minimum temperature due to its superior performance and consistency.

The results in Table 6 highlight the performance of various models for predicting average rainfall. The RFR achieved a MAE of 13.89 and an R^2 of 0.91, indicating a reasonable fit, but it exhibited a high MAE relative to the other models. LR demonstrated excellent performance with an MAE of 3.65 and an R^2 of 0.99, showing it closely approximates the actual values. Similarly, the LassoCV model also performed well with an MAE of 4.30 and R^2 of 0.99. The KNN Regressor had a higher MAE of 12.73 and R^2 of 0.92, indicating less performance than the linear models. The Huber Regressor and ANN model produced MAEs of 5.29 and 6.66, respectively, with R^2 values of 0.99 and 0.98, suggesting they also provided good predictions. However, the Ensemble Model, yielded the best results with the lowest MAE of 3.46 and an R^2 of 0.99, making it the most reliable model for average rainfall prediction in this analysis.

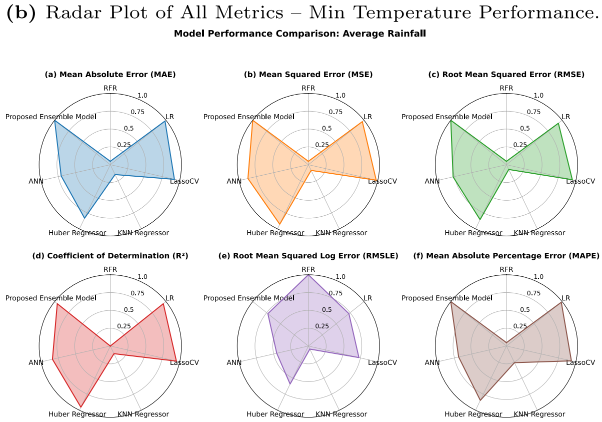
Figure 3 provides a clear, visual comparison of how different models performed in predicting key climate indicators–maximum temperature, minimum temperature, and average rainfall. All metric values were normalized for clearer visualization. Figure 3a, 3b, and 3ccorrespond to Tables 4, 5, and 6, respectively. For maximum temperature Fig. 3a, standard models showed weaker results, while the Huber Regressor did well except in R^2 and RMSLE. Still, our proposed model outperformed all others across the board. In minimum temperature predictions Fig. 3b, standard models performed slightly better, but again, our model showed stronger overall results, with only a small drop in RMSLE. For average rainfall 3c, some models did well in certain metrics, like RFR in RMSLE, but overall, our



(a) Radar Plot of All Metrics – Max Temperature Performance.

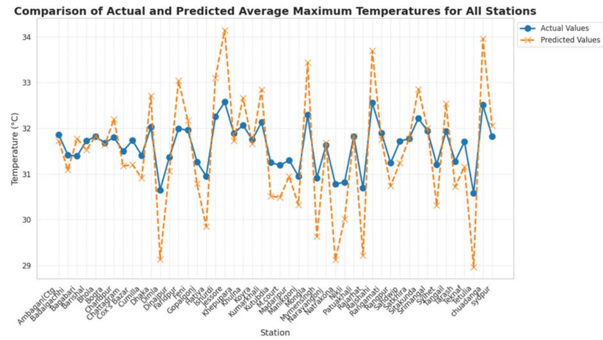


(b) Radar Plot of All Metrics – Min Temperature Performance.

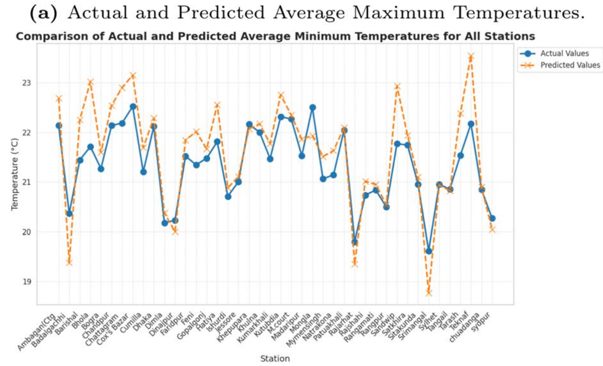


(c) Radar Plot of All Metrics – Average Rainfall Performance.

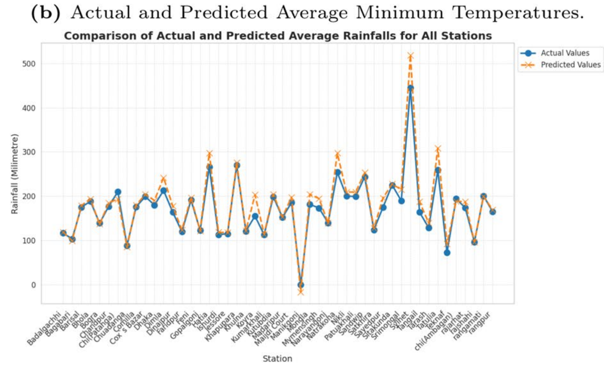
Fig. 3 Comparative radar analysis of predictive model performance across key climate indicators



(a) Actual and Predicted Average Maximum Temperatures.



(b) Actual and Predicted Average Minimum Temperatures.



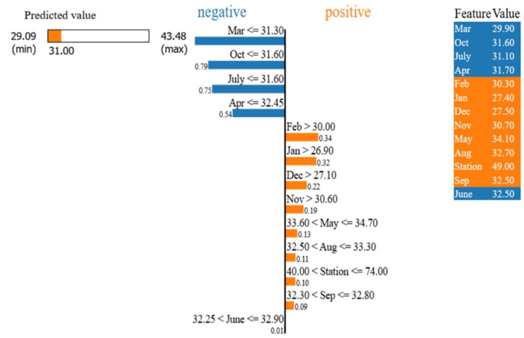
(c) Actual and Predicted Average Rainfall.

Fig. 4 Actual and predicted average maximum, minimum temperatures, and rainfall across various stations

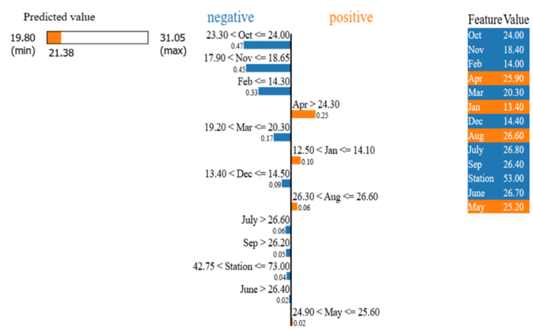
proposed approach delivered the most consistent and accurate performance.

Figure 4 presents a comparison of the actual and predicted values for standard average maximum and minimum temperatures and rainfall, based on the tested 2023 data from the Bangladesh Statistical Yearbook across different stations. Figure 4adisplays the comparison between the actual and predicted average maximum temperatures across various stations. The blue line with circular markers represents the actual temperatures, while the orange dashed line with 'X' markers indicates the predicted values, ranging from 29 to 34 °C. Overall, the predicted values closely follow the actual temperatures, although notable deviations occur at certain stations, such as Narail, Dhaka, and Rangpur, where significant discrepancies are observed. Despite these mismatches, the model effectively captures the general trend of temperature changes across most stations.

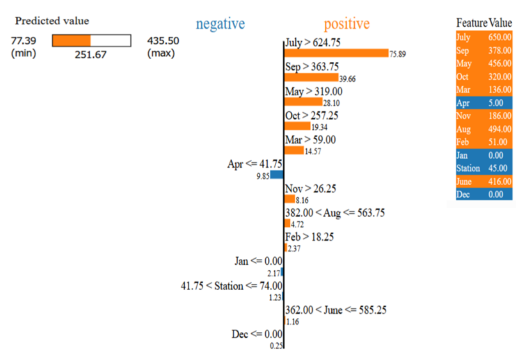
Figure 4bpresents a comparison of the actual and predicted average minimum temperatures for the same stations, with temperatures ranging from 19 to 2 °C. Notable discrepancies are evident, particularly at stations like Mymensingh and Tangail, where the predicted temperatures either overshoot or undershoot the actual values. Conversely, at locations such as Cox's Bazar and Faridpur, the predictions closely align with actual observations. Overall, the model



(a) LIME Tabular Explainer Plot for Average Maximum Temperature.



(b) LIME Tabular Explainer Plot for Average Minimum Temperature.



(c) LIME Tabular Explainer Plot for Average Rainfall.

Fig. 5 LIME tabular explainer for individual predictions of the proposed ensemble model

performs reasonably well, though it exhibits greater deviations in the lower temperature range at certain stations.

Figure 4ccompares the actual and predicted rainfall averages in millimeters across all stations, with values ranging from 100 mm to over 500 mm. The blue line represents the actual rainfall, while the orange dashed line indicates the predicted values. Generally, the predictions align closely with the actual measurements; however, notable overestimations occur at stations like Rangamati and Cox's Bazar. Overall, the model demonstrates better predictive performance for rainfall compared to the temperature predictions, suggesting it effectively captures rainfall trends across various stations.

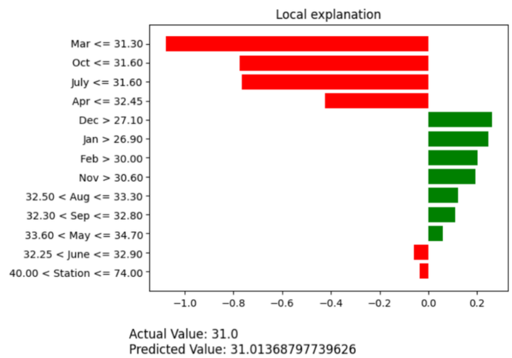
Using LIME to Interpret Ensemble Model's Predictions

This section presents an analysis of LIME's tabular explainer plots and feature importance plots, providing valuable insights into the model's predictions. Below is a detailed overview of the LIME plots for all dataset features.

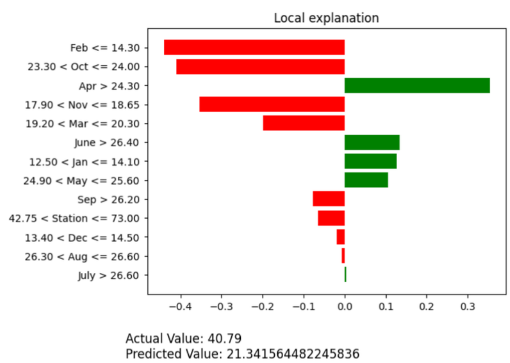
Figure 5 presents the LIME Tabular Explainer, illustrating individual predictions of the proposed ensemble model for both temperature and rainfall datasets. In Fig. 5a, the predicted maximum temperature is 31.05°C, with a range from a minimum of 28.83°C to a maximum of 42.79°C. Negative contributors include temperatures in March, July, October, and April, which fall below critical thresholds (31.30°C, 31.60°C, and 32.45°C). Notably, March temperatures at or below 31.30°C contribute significantly negatively (−0.94), as do July temperatures at or below 31.60°C (−0.79). Conversely, higher January temperatures above 26.90°C (0.37) and February temperatures exceeding 30.00°C (0.32) positively influence the prediction. Key feature values are 29.90°C for March, 31.10°C for July, and 49.00 at the station.

In Fig. 5b, the predicted minimum temperature is 21.38°C, with a range from 19.80°C to 31.05°C. Significant negative contributors include October temperatures between 23.30°C and 24.00°C (−0.47) and November temperatures between 17.90°C and 18.65°C (−0.45). Additionally, February temperatures below 14.30°C and March temperatures under 20.30°C further diminish the prediction. On the positive side, April temperatures above 24.30°C (0.25) and moderate January ($12.50°C < \text{Jan} < 14.10°C$) and August ($26.30°C < \text{Aug} < 26.60°C$) temperatures enhance the prediction. Key feature values are 24.00°C for October, 18.40°C for November, and 25.90°C for April.

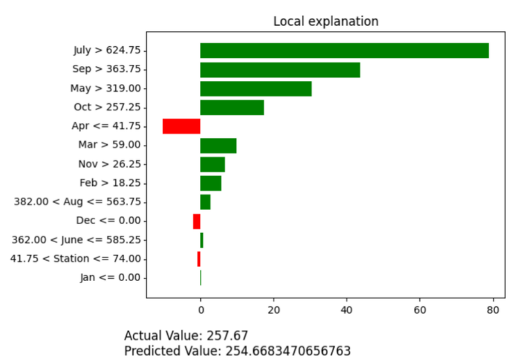
In Fig. 5c, the predicted rainfall is 252.28 mm, ranging from 74.39 mm to 474.76 mm. Key positive contributors include July rainfall exceeding 624.75 mm (adding 78.17 mm) and September rainfall above 363.75 mm (contributing 38.93 mm). Additionally, May and October rainfall values above 319.00 mm and 257.25 mm, respectively, also positively influence the prediction. Negative contributors include April rainfall of 41.75 mm or less (reducing the prediction by 11.41 mm) and low station rainfall between 41.75 mm and 74.00 mm (decreasing it by 2.72 mm). Significant feature values are 650.00 mm for July, 378.00 mm for September, and 456.00 mm for May.



(a) LIME Feature Importance Plot for Average Maximum Temperature



(b) LIME Feature Importance Plot for Average Minimum Temperature.



(c) LIME Feature Importance Plot for Average Rainfall.

Fig. 6 LIME feature importance plot for individual predictions of the proposed ensemble model

Figure 6 illustrates the LIME Feature Importance Plot, highlighting the individual predictions of the proposed ensemble model for all temperature and rainfall datasets. In Fig. 6a, the maximum temperature prediction, with an actual value of 31.0°C and a predicted value of 31.01°C, reveals key negative influences from March at −1.1, October at −0.9, and July at −0.9, along with April contributing −0.4. Conversely, December and January each positively impact the prediction with a contribution of +0.3. February and November also contribute positively, albeit with smaller effects of +0.2 each.

Table 7 Comparison of proposed method with existing works

No.	Reference	Year	Model(s)	Performance metric(s)
1	Liyew et al. [35]	2021	MLR, RF, XGBoost	MAE = 3.58, RMSE = 7.85
2	Milad et al. [38]	2021	Bi-LSTM, CNN, LSTM, GRU	R^2 = 0.9555
3	Kanani et al. [31]	2023	XGBoost, RF, SVM, MLR, LSTM	Accuracy = 92.2%, MAE = 11.7%, precision = 95.6%, F1 score = 91.9%, and recall 91.1%
4	Azeem et al. [9]	2023	SVM, Neural Networks, ANN, Voting Ensemble	Accuracy = 86.5%
5	Akram et al. [4]	2024	NB, ANN, LogitBoost, RIPPER, DS, AdaBoost, RF	Accuracy = 83.2%
6	**Our Study (Proposed)**	**2025**	**PCA-IG, Ensemble Model, and XAI**	**Max Temperature (MAE = 0.08, RMSE = 0.15, R^2= 1.00), Min Temperature (MAE = 0.39, RMSE = 0.53, R^2= 0.73), Average Rainfall (MAE = 3.46, RMSE = 5.15, R^2 = 0.99)**

In Fig. 6b, the minimum temperature prediction, with an actual value of 40.79°C and a predicted value of 21.34°C, shows significant negative influences from February ($\leq$ 14.30°C) at −0.4, and both October and April at approximately −0.3. November and March also negatively impact the prediction, with contributions of −0.3 and −0.2, respectively. On the positive side, June ($>$ 26.40°C) adds +0.2, while January and May contribute smaller positive values of +0.2 and +0.1. These findings indicate that the model anticipates higher minimum temperatures in certain months while penalizing others, resulting in an overall lower predicted value.

Lastly, in Fig. 6c, the rainfall prediction shows an actual value of 257.67 mm and a predicted value of 254.67 mm. The main positive contributors include July ($>$ 624.75 mm), which significantly adds +80, and September ($>$ 363.75 mm), contributing +40. Additionally, May and October increase the prediction by +30 and +20, respectively. Conversely, April ($\leq$ 41.75 mm) detracts – 10, and December ($\leq$ 0.00 mm) has a minimal negative impact of – 2. These findings suggest that the model correlates heavy rainfall with July and September, while lighter rainfall in April and December reduces the overall prediction.

Benchmarking Against State-of-the-Art Models

In this section, we compare our proposed model with prior studies within the same domain [4, 9, 35, 38] that utilized temperature or rainfall data for predictive analysis using various techniques such as MLR, RF, XGBoost, Bi-LSTM, CNN, LSTM, GRU, and others. However, many of these studies did not incorporate robust feature selection strategies XAI techniques, limiting their interpretability and predictive reliability [31, 31, 35]. In contrast, our study introduces a PCA-IG-based feature selection method, a novel ensemble learning model, and integrated XAI approaches to more accurately identify influential factors. As a result, our model demonstrates superior performance across all evaluation metrics when compared to traditional and previously published models. A comprehensive comparison is presented in Table 7, highlighting the state-of-the-art advancement offered by our approach.

Discussion

Our study brings fresh insights into Bangladesh's climate patterns by focusing on detailed monthly averages of temperature and rainfall. Many earlier works, as shown in Table 1, did not explore these local climate factors in depth. We also found that previous studies often missed the opportunity to apply proper feature selection or XAI techniques [4, 9, 13, 19, 38]. By addressing these gaps, our approach offers a clearer and more informed understanding of climate trends and model behavior. Our PCA-IG based feature importance analysis clearly shows that certain months play a bigger role in predicting climate patterns in Bangladesh. For maximum temperature, April and February stood out the most. For minimum temperature, September and June were key. When it comes to rainfall, months like June, April, and August had the strongest impact. On the other hand, months such as November and October contributed very little across all datasets. Interestingly, the station (location) mattered more for rainfall predictions. These findings highlight the importance of focusing on specific months to improve climate prediction models.

In our results section, we tested several ML and DL models across three climate datasets: average maximum temperature, average minimum temperature, and average rainfall. Among all the models, our proposed ensemble model consistently delivered the best performance. For the maximum temperature dataset, the model achieved an impressive R^2 of 1.00 and a very low MAE of 0.08. In the minimum temperature dataset, it reached an R^2 of 0.79 with an MAE of 0.39. For rainfall, the model performed strongly as well, with an R^2 of 0.99 and an MAE of 3.46. Our proposed model, when compared with traditional models and those from previous studies, consistently outperforms them in terms of overall performance. These results highlight the robustness and reliability of our hybrid approach in capturing complex climate patterns.

Our model shows strong predictive performance when tested on 2023 climate data from various stations across Bangladesh. For maximum temperature, it closely aligned with actual values in most areas, though deviations were noted in stations like Narail, Dhaka, and Rangpur. In the case of minimum temperature, the model performed well overall, but some mismatches appeared in Mymensingh and Tangail, while predictions for Cox's Bazar and Faridpur were highly accurate. For rainfall, the model demonstrated excellent trend detection across stations, with close alignment in most areas and slight overestimations in Rangamati and Cox's Bazar. These results highlight the model's reliability and effectiveness in forecasting climate patterns, which can support better decision-making in agriculture and disaster preparedness.

By applying XAI techniques like LIME (both tabular and feature importance plots), we gained meaningful insights into the factors influencing our model's predictions. In the tabular analysis, for maximum temperature, lower values in March and July negatively affected predictions, whereas higher temperatures in January and February had a strong positive influence. In minimum temperature, cooler October and November lowered predictions, whereas warmer April and January had positive effects. For rainfall, high values in July and September strongly influenced the model, while low rainfall in April reduced prediction scores. LIME feature importance plot analysis showed that for maximum temperature, March, October, and July negatively influenced predictions, while December and January had positive effects. In minimum temperature predictions, February, October, and April lowered values, whereas June and January boosted them. For rainfall, higher values in July and September drove predictions up, while low rainfall in April and December reduced them–highlighting how seasonal patterns influence model outcomes across all datasets.

We compared our approach with previous works and found that our model consistently delivered stronger performance [31, 35, 38]. By integrating feature selection, an ensemble learning approach, and XAI techniques, we uncovered valuable new insights. Notably, we observed regional variations in prediction accuracy across all datasets, reflecting diverse climatic patterns. These findings underscore the importance of developing region-specific climate models and policies to enable more effective and localized adaptation strategies.

Conclusion and Future Work

This study demonstrates the effectiveness of a hybrid ML and DL approach for predicting climate change impacts on temperature and rainfall in Bangladesh. We developed an ensemble model that outperforms traditional forecasting methods by incorporating both maximum and minimum temperature estimates alongside rainfall data. To enhance interpretability, we employed XAI techniques such as LIME to identify the most influential features driving the predictions. Our model not only captures the overall trends in climatic variables but also addresses key gaps highlighted in previous studies, including the absence of feature importance analysis and the lack of an integrated framework combining temperature and rainfall data.

Looking ahead, future studies can make this ensemble approach even stronger by including more meteorological variables like humidity, wind speed, soil moisture, and evapotranspiration. Considering large-scale climate patterns such as ENSO (El Niño–Southern Oscillation) could also help capture deeper climate influences, especially across South Asia. Expanding the dataset to cover more diverse regions would improve how well the model works in different conditions. To make the model's predictions easier to understand, more advanced XAI tools like SHAPASH, DeepSHAP could be used. Lastly, exploring new technologies like quantum machine learning might bring exciting improvements, offering faster, smarter ways to forecast climate changes and support better decision-making.

Data Availability The codes and data employed in this study can be found in the following GitHub repository.https://github.com/ShahriarAyon63/temperature_rainfall

Declarations

Conflict of Interest The authors declare no conflict of interest.

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